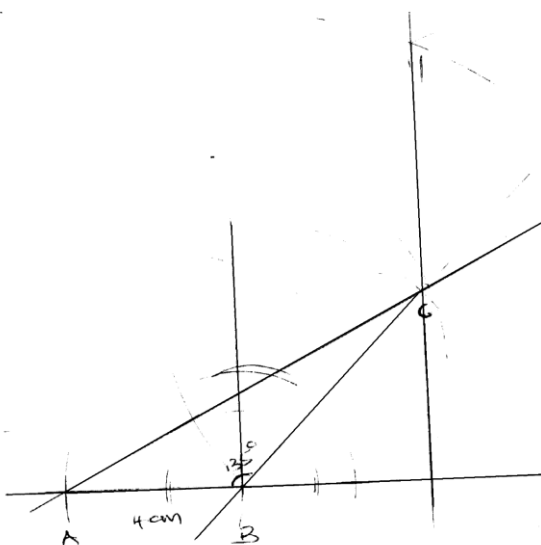
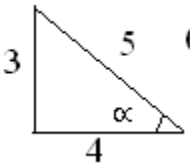


Kenya Certificate of Secondary Examination (KCSE)

121/1

MATHEMATICS**PAPER 2****MARKING SCHEME**

	Questions	Marks	Remarks
1.	$\frac{\frac{4}{11} \times \frac{14}{20}}{\frac{10}{3} \div \frac{11}{10}}$ $= \frac{\frac{4}{55}}{\frac{10}{3} \times \frac{11}{10}}$ $= \frac{4}{55} \times \frac{33}{100}$ $= \frac{3}{125}$	 B1 B1 A1	
2.	$500 \times 84.15 - 289850$ $= 130,900$ But 100 yens $\equiv$ 65 – 45 ? 130900 $= 200000 \text{ yens}$	 M1 M1 A1	
3.	Gradient L1 = $\frac{-4+2}{3-1} = \frac{-2}{2} = -1$ → Gradient L2 = 1 Mid- point AB = (2, -3) $\therefore \frac{y+3}{x-2} = 1$ → $y + 3 = x - 2$ $\therefore -x + y + 5 = 0$	 B1 B1 M1 A1	
4.	$\Pi dh = 1980$ $\frac{22}{7} \times 42 \times h$ $h = 15\text{cm}$ $\text{Vol} = \frac{22}{7} \times 21 \times 21 \times 15$ $= 20790\text{cm}^3$ $= 20.79\text{li}$	 M1 M1 A1	
5.	$\frac{3a+2}{4} \leq \frac{2a+3}{5}$ $\rightarrow 5(3a+2) \leq 4(2a+3)$ $15a + 10 \leq 8a + 12$ $7a \leq 2$	 M1	

	$a \leq \frac{2}{7}$ $\frac{2a+3}{5} \leq \frac{4a+15}{6}$ $6(2a+3) \leq 5(4a+15)$ $12a+18 \leq 20a+75$ $-8a \leq 57$ $A \geq -\frac{57}{8} \Rightarrow -\frac{57}{8} \leq a \leq \frac{2}{7}$ Intergal values -7, -6, -5, -4, -3, -2, -1, & 0	M1 M1 A1	
6.			
7.	$80 + (n-1)128 = 180n - 360 \quad \text{or its equivalent}$ $\rightarrow 80 + 128n - 128 = 180n - 360$ $-52n = -312$ $n = 6$	M1 M1 A1	
8.	 $0.75 = 3/4$ $\rightarrow \cos \alpha = \frac{4}{5}$	M1 M1 A1	



	$\cos(180 - \alpha) = -\frac{4}{5}$		
9.	$\frac{x^4 - y^4}{x(x^2 - y^2)}$ $\frac{(x^2 + y^2)(x^2 - y^2)}{x(x^2 - y^2)}$ $\frac{x^2 + y^2}{x}$	M1 M1 A1	

10.	Let Area c x y = x L.S. F = $\frac{4}{12} = \frac{1}{3}$ A.S.F = $\frac{1}{9}$ $\rightarrow \frac{1}{9} = \frac{x}{x+36}$ $\rightarrow 9x = x + 36$ $8x = 36$ $X = 4.5\text{cm}^2$	M1 M1 A1	
11.	DIAGRAM XO = (r-4)cm OP = rcm XP = 10cm $\rightarrow r^2 = 100 + (r-4)^2$ $r^2 = 100 + r^2 - 8r + 16$ $\rightarrow 8r = 116$ $r = 14.5\text{cm}$	M1 M1 A1	
12.	i) 19, 171, 311 ii) 300	B1 B1	
13.	$\left(\frac{1}{4}\right)^{2X} = \left(\frac{1}{32}\right)^{3X-4}$ $\left(\frac{1}{2}\right)^{4X} = \left(\frac{1}{2}\right)^{15X-20}$ $\rightarrow 4x = 15x - 20$ $-11x = -20$ $X = \frac{20}{11}$	M1 M1 A1	
14.	$450000 \equiv 100\%$ $? \equiv 93\%$ Cash price = 418500 shillings But 418500 $\equiv 113\%$ $? \equiv 100\%$		



	= 370353.9823 shillings	A1	
15.	$(3.375 \times 1/10)^3 = 0.038441$ $(3.375 \times 100)^{1/2} = 18.372$ $= \frac{3}{-18.333559}$ $(-3) \times \frac{1}{18.33} = (-3) \times \frac{1}{1.833} \times 10$ $= (-3) \times (0.5455) \times 1/10$ $= -0.16365$	A1	
16.	$\begin{array}{r} -\frac{14}{1} + \frac{4}{3} \\ -6 + -20 \\ -38 \\ = \frac{3}{-26} \\ = \frac{-38}{3} \times \frac{1}{-26} \\ = \frac{38}{3} \times \frac{1}{26} = \frac{19}{39} \end{array}$	 B1 B1 A1	
17.	a) Checkpoint $\begin{array}{cc} & \\ \text{Matatu} & \text{Bus} \\ 10.15\text{am} & 10.15\text{am} \end{array}$ Distance between matatu and bus = distance covered by bus for 15min $= 99\text{km/h} \times \frac{15}{60} = 24.75\text{km}$ Time taken for matatu to catch up with bus $= \frac{\text{Distance between}}{\text{Relative speed}}$ $= \frac{24.75\text{km}}{33\text{km/h}} = 0.75\text{hrs} = 45\text{min}$ → Overtaking will take place at 10.15am + 45min = 11.00am b) Time taken = $\frac{\text{Distance}}{\text{R.S}}$ $= \frac{(150 + 240 + 100)}{252\text{km/h}}$ But 252km/h = 70m/s $T = \frac{490\text{m}}{70\text{m/s}} = 7\text{sec s}$	 M1 M1 M1 M1 M1 A1 M1M1 M1 A1	

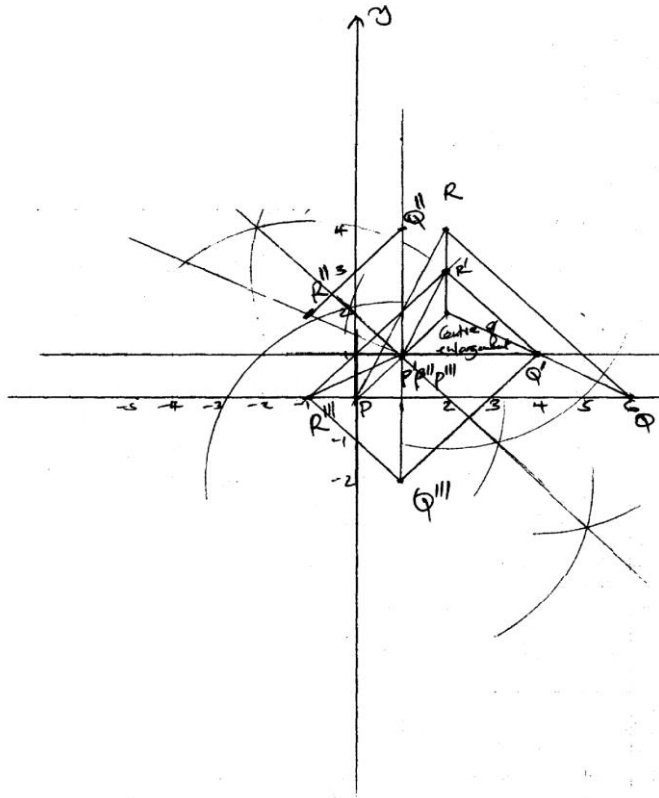
18.

- a) Draw PQR
 b) $P'(1, 1)$ $Q'(4, 1)$ $R'(2, 3)$
 c) $P''(1, 1)$ $Q''(1, 4)$ $R''(-1, 2)$
 d) $P'''(1, 1)$ $Q'''(1, -2)$ $R'''(-1, 0)$
 e) Reflection through the mirror line $y = -x + 2$

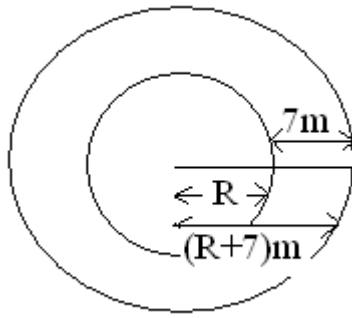
Gradient = -1

$$\rightarrow \frac{y-1}{x-1} = -1$$

$$y - 1 = -x + 1$$



19.

a) let radius lawn R mradius of lawn r path $= (R+7)m$ Area of the lawn $= \pi R^2 m^2$ Area of the lawn + path $= \pi(R+7)^2$

$$= \pi(R^2 + 14R + 49)$$

$$= (R^2 + 14R + 49) \pi m^2$$

Area of the path $= (R^2 + 14R + 49) \pi R^2$

$$= \pi R^2 + 14\pi R + 49\pi - \pi R^2$$

$$= 14\pi R + 49\pi = (14R + 49)\pi$$

Area of the path also equal to 21% of area of lawn

$$\rightarrow 21\% \text{ of } \pi R^2$$

$$= 0.21\pi R^2$$

$$\rightarrow 0.21\pi R^2 = (14R + 49)\pi$$

$$0.21R^2 = 14R + 49$$

$$0.21R^2 - 14R - 49 = 0$$

$$21R^2 - 1400R - 4900 = 0$$

$$3R^2 - 200R - 700 = 0$$

$$R = \frac{200 \pm \sqrt{(-200)^2 - 4(3)(-700)}}{6}$$

$$= \frac{200 \pm 220}{6} \Rightarrow \frac{-20}{6} \text{ or } \frac{420}{6}$$

$$= \frac{-20}{6} \text{ impossible}$$

$$= 70$$

Hence $R = 70$ m

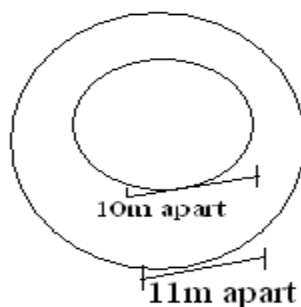
M1

B1

M1

M1

b)



Perimeter of inner fence

$$= 2\pi R$$

$$= 2 \times 3.14 \times 70 = 439.88 \text{ m}$$

$$\text{No of post } \frac{439.88}{10} = 43.988 = 44 \text{ posts}$$

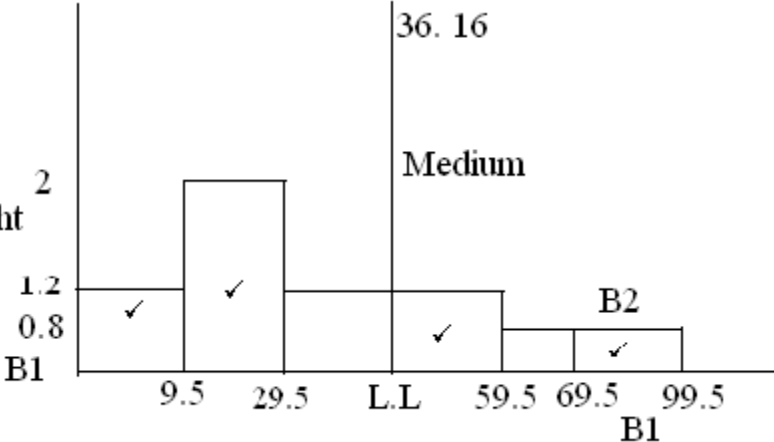
Perimeter of outer fence

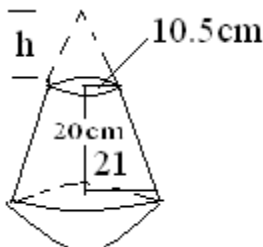
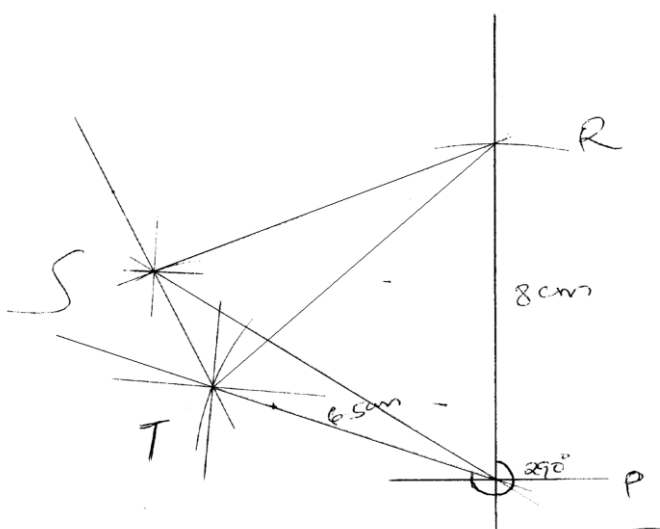
M1

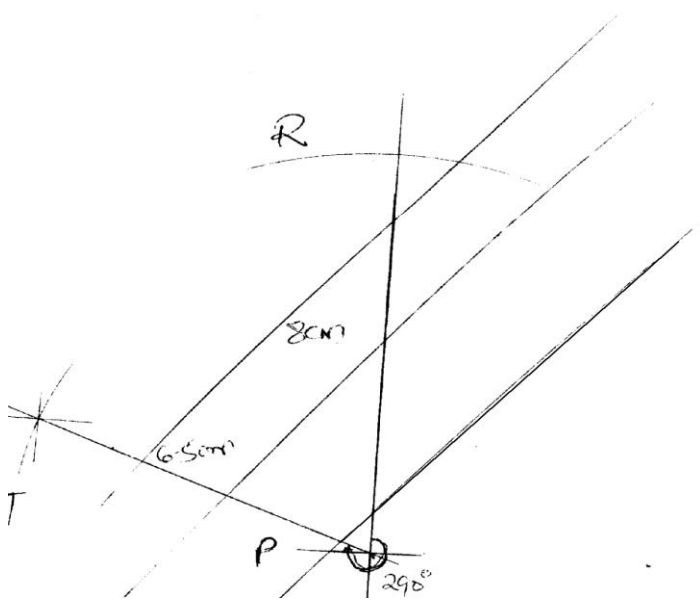


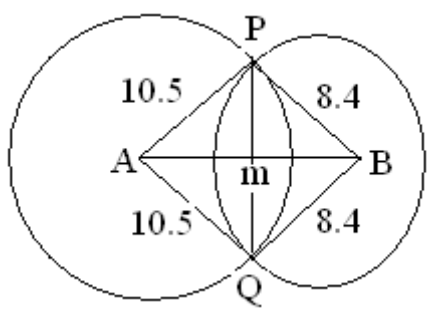
	$= 2 \times 3.14 \times 77 = 483.868\text{m}$ $\text{No of posts} = \frac{483.868}{11} = 43.988$ $= 44 \text{ posts}$ $\text{Total number of posts} = 44 + 44 = 88\text{posts}$ c) Total cost $88 \times 103 - 1 = \text{Sh. } 9240$	M1 M1 A1 A1	
20.	a) $v = at^2 + bt \text{ m/s}$ at $t = 1\text{sec}$ $v = 2\text{m/s}$ $a + b = 2 \dots\dots\dots(i)$ at rest $v = 0, t = 45$ $(4.5)^2 a + 45b = 0$ $20.25 a + 4.5 b = 0 \dots\dots\dots(ii)$ Solving (i) and (ii) simultaneously $a = 2 - b$ $20.25 (2 - b) + 4.5b = 0$ $40.5 + 4.5 b = 0$ $b = \frac{-40.5}{4.5} = -9$ $\rightarrow a = 2 - (-9) = 11$ $v = 11t^2 - 9tm/s$ b) i) at rest, $v = 0$ $11t^2 - 9t = 0$ $(11t - 9)t = 0$ $\rightarrow t=0 \text{ or } 11t = 9 \rightarrow t = \frac{9}{11}$ ii) $s = \int_0^4 (11t^2 - 9t)dt$ $\left[\frac{4}{3}t^3 - \frac{9}{2}t^2 + c \right]_0^4$ $\left(\frac{11}{3}(4)^3 - \frac{9}{2}(4)^2 + c \right) - \left(\frac{11}{3}(0)^3 - \frac{9}{2}(0)^2 + c \right)$ $(234.67 - 72 + c) - (c)$ $= 162.67 \text{ m}$ iii) at max velocity $a = 0$ $a = \frac{dv}{dt}$ $v = 11t^2 - 9t$ $a = 22t - 9$ $22t - 9 = 0$ $t = \frac{9}{22}$ $v = 11 \left(\frac{9}{22} \right)^2 - 9 \left(\frac{9}{22} \right)$ $= 11 \frac{81}{484} - \frac{81}{22}$	M1 M1 M1 A1 M1 A1 M1	

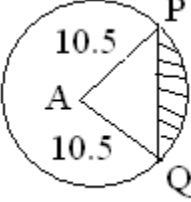
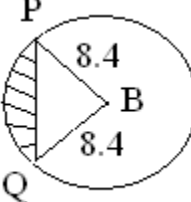


	$= \frac{81}{44} - \frac{81}{22} = \frac{81-162}{44}$ $= \frac{-81}{44} m/s \text{ deceleration}$						A1		
21.	Marks	0-10	10-30	30-60	60-70	70-100	B2		
	Frequency	12	40	36	8	24			
	Area of rec	12	40	36	8	24			
	Height of rec	12	2	1.2	0.8	0.8			
	L.L	9.5	29.5	59.5	69.5	99.5			
	c.f	12	52	88	96	120			
ii)									
									
ii) Medium position = $\frac{N+1}{2} = \frac{121}{2} = 60.5$ in the class 30- 60								B1	
iii) $\frac{1}{2}$ total area = $\frac{120}{2} units^2 = 60 units^2$ $= 12 + 40 + 8 = 60 units^2$								M1	
But $\frac{8}{1-2} = 6.66 + 29.5 = 36.16$								A1	

22.	 a) $\frac{h}{20+h} = \frac{10.5}{21}$ $21h = 10.5h + 210$ $10.5h = 210$ $h = 20\text{cm}$ $\rightarrow H = 40\text{cm}$ $L^2 = 21^2 + 40^2$ $L^2 = 441 + 1600$ $L = 45.17742799\text{cm}$ b) $L_2 = 20^2 + 10.5^2$ $L_2 = 400 + 110.25$ $L = 22.588714\text{cm}$ $\rightarrow$ slant length of frustum $= 45.17742799 - 22.588714$ $= 22.58871399$ c) TOTAL S.A = $\pi r^2 + (\pi RL - \pi rL) + 2\pi R^2$ $=$ $\left(\frac{22}{7} \times 10.5 \times 10.5\right) + \left[\left(\frac{22}{7} \times 21 \times 45.17742799\right) - \left(\frac{22}{7} \times 10.5 \times 22.588714\right)\right]$ $+ 2 \times \frac{22}{7} \times 21 \times 21$ $= 346.5 + 2981.7104 - 745.427562 + 2772$ $= 5354.782685\text{cm}^2$	M1 M1 A1 B1 B1 M1 M1 M1 A1	
23.			

	 a) 84km, 042° b) 80km, 065° c) 120°		
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24.	a)  $\angle PAQ = \angle PAM + \angle QAM$ $\angle PAM = \sin^{-1} \frac{4.5}{10.5} = 0.4286$ $\sin^{-1}(0.4286) = 25.38^\circ$ $\angle QAM = \angle PAM = 25.38$ $\rightarrow \angle LAP = 25.38 \times 2 = 50.76$ b) $\angle PBQ = \angle PBM + \angle QBM$ $\angle PBM = \sin^{-1} \frac{45}{84} = 0.5357$ $\sin^{-1}(0.5357) = 32.39^\circ$ $\angle PBM = \angle QBM = 32.39^\circ$ $\angle PBQ = 32.39^\circ \times 2 = 64.78$ c)i)	M1 A1 M1 A1 M1 B1 M1M1 B1	
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	  area of segment = area of a section – area of D Taking (i) $= \left[\frac{50.76}{360} \times 3.14 \times (10.5)^2 \right] - \left[\frac{1}{2} \times 10.5 \times 10.5 \times \sin 50.76 \right]$ $= 48.84 - 42.69 = 6.15 \text{ cm}^2$ Taking (ii) $= \left[\frac{64.78}{360} \times 3.142 \times (8.4)^2 \right] - \left[\frac{1}{2} \times 8.4 \times 8.4 \times \sin 64.78 \right]$ $= 39.89 - 31.92 = 7.97 \text{ cm}^2$ $= (6.15 + 7.97) \text{ cm}^2 = 14.12 \text{ cm}^2$	A1	